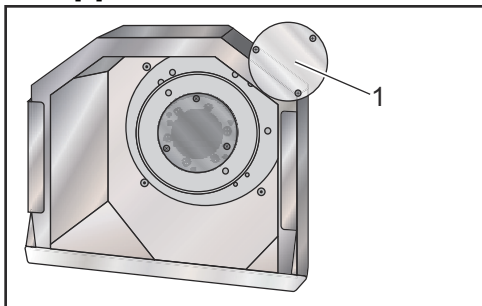


4.8.2 Drawtube Cover Plate

Before using the Bar Feeder,

F4.16: Draw Tube Cover Plate [1].



1. Remove the cover plate [1] at the far end of the drawtube.
2. Replace the cover plate anytime bar stock is not being fed automatically.

4.9 Tooling

This section describes tool management in the Haas control: commanding tool changes, loading tools into holders, and Advanced Tool Management.

4.9.1 Advanced Tool Management Introduction

Advanced Tool Management (ATM) lets you set up groups of duplicate tools for the same or a series of jobs.

ATM classifies duplicate or backup tools into specific groups. In your program, you specify a group of tools instead of a single tool. ATM tracks the tool use in each tool group and compares it to your defined limits. When a tool reaches a limit, the control considers it “expired.” The next time your program calls that tool group, the control chooses a non-expired tool from the group.

When a tool expires:

- The beacon will flash.
- ATM puts the expired tool in the **EXP** group
- Tool groups that contain the tool appear with a red background.

To use ATM, press **[CURRENT COMMANDS]**, and then select ATM in the tabbed menu. The ATM window has two sections: **Allowed Limits** and **Tool Data**.

F4.17: Advanced Tool Management Window: [1] Active window label, [2] Allowed Limits window, [3] Tool Group window, [4] Tool Data window

Current Commands

TimersMacro VarsActive CodesATMCalculatorMediaOscilloscope

F4 To Switch BoxesAllowed LimitsActive Tool: 1

Group	Expired Count	Tool Order	Holes Limit	Usage Limit	Life Warn %	Load Limit	Expired Action	Feed Limit	Total Time Limit
All	-	-	-	-	-	-	-	-	-
Expired	0	-	-	-	-	-	-	-	-
No Group	-	-	-	-	-	-	-	-	-
Add Group	-	-	-	-	-	-	-	-	-

Tool Data For Group: All

Tool	Offset	Life	Holes Count	Usage Count	Usage Limit	Max Load %	Load Limit %	Feed Time	Total Time
1	1	100%	0	0	0	0%	0%	0:01:07	7:10:07
2	2	100%	0	0	0	0%	0%	0:00:00	0:00:20
3	3	100%	0	0	0	0%	0%	0:00:00	0:00:00
4	4	100%	0	0	0	0%	0%	0:00:00	0:00:00
5	5	100%	0	0	0	0%	0%	0:00:00	0:00:00
6	6	100%	0	0	0	0%	0%	0:00:00	0:00:00

INSERT Add Group

Allowed Limits

This table gives data about all of the current tool groups, including default groups and user-specified groups. **ALL** is a default group that lists all of the tools in the system. **EXP** is a default group that lists all of the tools that are expired. The last row in the table shows all of the tools that are not assigned to tool groups. Use the cursor arrow keys or **[END]** to move the cursor to the row and see these tools.

For each tool group in the **ALLOWED LIMITS** table, you define limits that determine when a tool expires. The limits apply to all tools assigned to this group. These limits affect every tool in the group.

The columns in the **ALLOWED LIMITS** table are:

- GROUP** - Displays the tool group's ID number. this is the number you use to specify the tool group in a program.
- EXP #** - Tells you how many tools in the group are expired. If you highlight the **ALL** row, you see a list of all of the expired tools in all groups.
- ORDER** - Specifies the tool to use first. If you select **ORDERED**, ATM uses the tools in tool number order. You can also have ATM automatically use the **NEWEST** or **OLDEST** tool in the group.
- USAGE** - The maximum number of times the control can use a tool before it expires.
- HOLES** - The maximum number of holes a tool is allowed to drill before it expires.

- **WARN** - The minimum value for tool life remaining in the group before the control gives a warning message.
- **LOAD** - The allowed load limit for tools in the group before the control does the **ACTION** that the next column specifies.
- **ACTION** - The automatic action when a tool reaches its maximum tool load percentage. Highlight the tool action box to change and press **[ENTER]**. Use the **[UP]** and **[DOWN]** cursor keys to select the automatic action from the pull down menu (**ALARM, FEEDHOLD, BEEP, AUTOFEED, NEXT TOOL**).
- **FEED** - The total amount of time, in minutes, that the tool can be in a feed.
- **TOTAL TIME** - The total amount of time, in minutes, that the control can use a tool.

Tool Data

This table gives information about each tool in a tool group. To look at a group, highlight it in the **ALLOWED LIMITS** table, and then press **[F4]**.

- **TOOL#** - Shows the tool numbers used in the group.
- **LIFE** - The percentage of life left in a tool. This is calculated by the CNC control, using actual tool data and the allowed limits the operator entered for the group.
- **USAGE** - The total number of times that a program has called the tool (number of tool changes).
- **HOLES** - The number of holes the tool has drilled/ tapped/ bored.
- **LOAD** - The maximum load, in percent, exerted on the tool.
- **LIMIT** - The maximum load allowed for the tool
- **FEED** - The amount of time, in minutes, the tool has been in a feed.
- **TOTAL** - The total amount of time, in minutes, the tool has been used.

Advanced Tool Management Macros

Advanced Tool Management (ATM) can use macros to obsolete a tool within a tool group. Macros 8001 to 8099 represent tools 1 through 99. You can set one of these macros to 1 to expire a tool. For example:

8001 = 1 (this expires tool 1)

8001 = 0 (this makes tool 1 available)

Macro variables 8500 - 8515 enable a G code program to obtain tool group information. When you specify a tool group ID number with macro 8500, the control returns the tool group information in macro variables #8501 through #8515. Refer to the variables #8500 - #8515 in the Macros chapter for the macro variable data label information.